SYED ANAS IMAM

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PROFESSIONAL SUMMARY

Computer Science undergraduate at Usman Institute of Technology and Trainee at MERL with proven hands-on experience engineering full-stack AI applications, processor architectures, and machine learning models. Built Career OS with FastAPI, Next.js, and Qwen LLM, and implemented 32-bit single-cycle RISC-V processor cores in Logisim and Chisel/Scala. Seeking software engineering, AI/ML, or systems internship roles.

TECHNICAL SKILLS

Languages: Python, Java, Scala, TypeScript, JavaScript, SQL, CSS
AI / Machine Learning: TensorFlow, LSTM, NumPy, Qwen LLM (DashScope API), Microsoft Copilot Studio
Web / Backend: FastAPI, Next.js, React, RESTful APIs, Pydantic, Docker
Systems / Hardware: RISC-V (RV32I ISA), Chisel HDL, Logisim, FIRRTL, Digital Logic Design
Developer Tools: Git, GitHub, VS Code, Vercel, Render, Swagger/OpenAPI, SQLite

PROFESSIONAL EXPERIENCE

Trainee | *Microelectronics Research Laboratory (MERL) | Karachi* *Aug 2026 – Present*

- Conducting instruction set architecture (ISA) modeling, instruction decoding, and RTL design for 32-bit RISC-V (RV32I) single-cycle processor cores using Chisel HDL (Scala) and Logisim.
- Simulating and validating datapath components including ALU operations, 32-register files, immediate generation, Harvard memory interfaces, and control logic for 9 instruction formats.

TECHNICAL PROJECTS

Career OS — AI-Powered Career Guidance Platform | *Full-Stack Web App* *Aug 2026 – Sep 2026*

Technologies: *FastAPI, Next.js, TypeScript, Qwen AI (DashScope), SQLite, Docker, Vercel, Render*

- Engineered a production-ready career platform featuring a FastAPI backend (20+ REST endpoints) and Next.js/TypeScript frontend, deployed and hosted on Render and Vercel.
- Integrated Qwen LLM via DashScope API with JSON schema enforcement and fallback logic for personalized career recommendations, mock interview generation, and interactive roadmaps.
- Built deterministic Pakistan Knowledge Engine (248 universities, 210 degrees, 22 tracks) and implemented an MCP server to expose structured career data directly to AI agents.

RISC-V Single-Cycle Processors (RV32I ISA) | *Computer Architecture* *Aug 2026 – Sep 2026*

Technologies: *Scala, Chisel HDL, FIRRTL, Logisim, RISC-V RV32I ISA, Digital Logic Design*

- Designed and implemented two 32-bit single-cycle RISC-V cores: one synthesizable in Chisel/Scala (generating FIRRTL) and one constructed from gate-level digital components in Logisim.
- Implemented Harvard architecture memory, 32-entry register file, 10-op ALU, immediate generator, branch comparator (6 conditions), and hierarchical control unit supporting 9 instruction formats.

LSTM Stock Price Predictor | *Deep Learning* *Academic Project*

Technologies: *Python, TensorFlow, NumPy, Time-Series Modeling*

- Built deep learning recurrent neural network using LSTM layers in TensorFlow to forecast stock trends from historical price sequences, incorporating data scaling and train/test pipelines.

Memory Card Game | *Software Engineering* *Academic Project*

Technologies: *Java, Object-Oriented Programming (OOP)*

- Developed interactive desktop card game applying core OOP principles (polymorphism, encapsulation, inheritance, abstraction) with dynamic card shuffling, score tracking, and integrated audio effects.

EDUCATION

BS in Computer Science | *Usman Institute of Technology, Karachi* *2025 – 2028 (Expected)*

- Relevant Coursework: Data Structures & Algorithms, Object-Oriented Programming, Computer Architecture, Digital Logic Design

A-Levels & O-Levels (Cambridge Board) | *St. Patrick's High School / Private Candidate, Karachi* *Completed*

CERTIFICATIONS & HONORS

- **Microsoft Certified: Level 1 Explorer — AI Agent** — Demonstrated proficiency in AI agent concepts and architectures.
- **AI Agent Developer (Microsoft Copilot Studio)** — Designed and published autonomous AI agent using Copilot Studio Agent Developer.
- **UIT Young Writers Award & Leadership** — Published Letter to Editor in The News International; House Captain (won inter-house sports & 6 track medals).